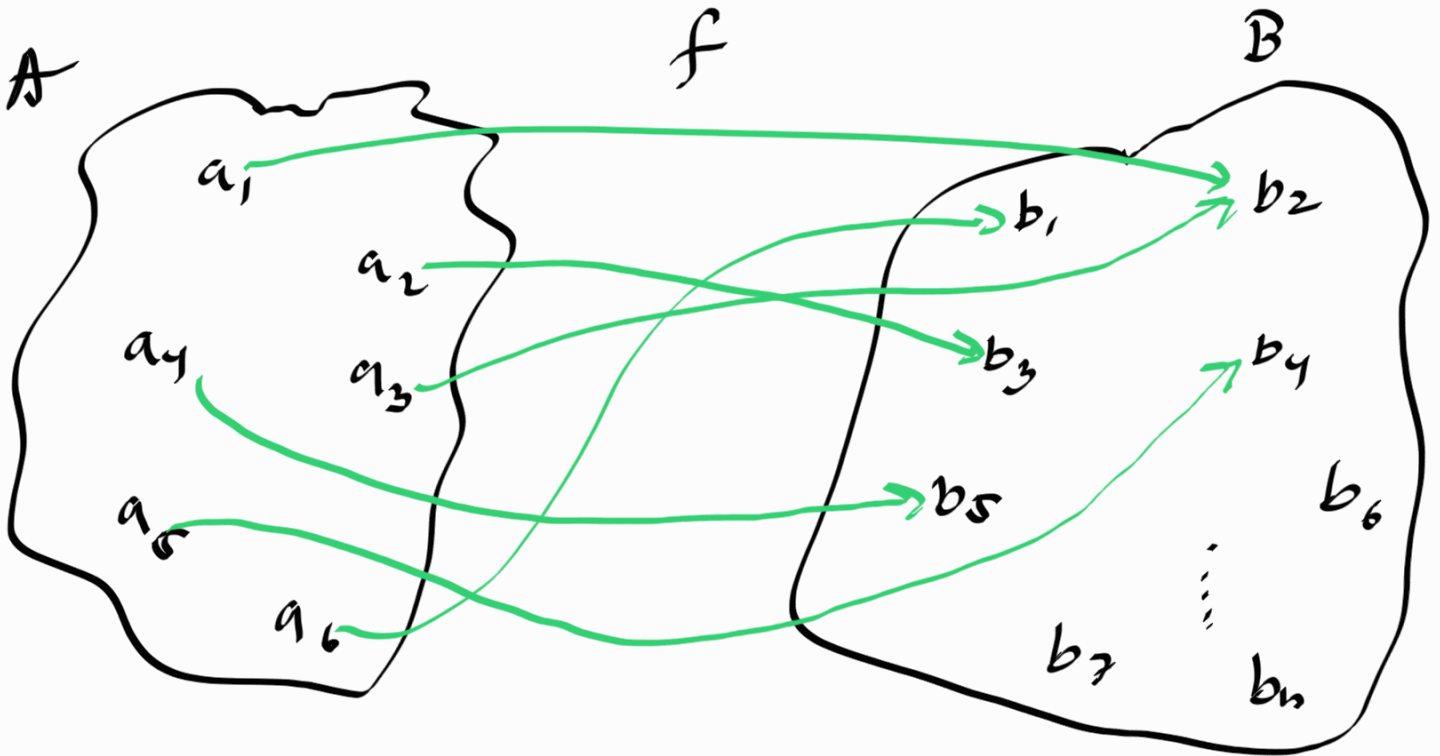


Functions

Def

Let A, B be sets. A function f from A to B , written as $f: A \rightarrow B$ assigns to each input value $a \in A$, a unique output value $b \in B$; the unique value that is assigned to " a " is denoted as $f(a)$. We sometimes say that f maps a to $f(a)$.



$$a \rightarrow f(a)$$

A, B can be the same set.

- predicates are functions

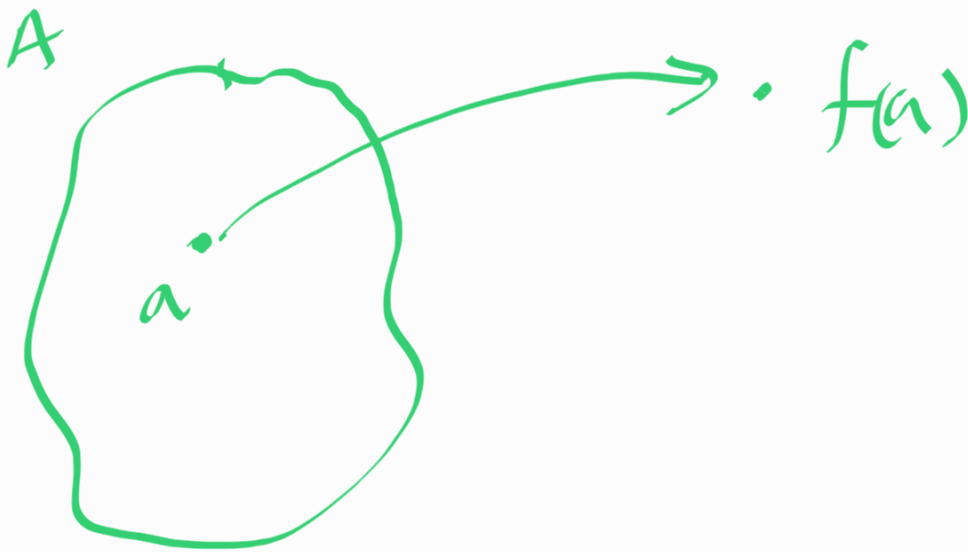
- is Even; $\mathbb{Z} \rightarrow \{T, F\}$

- $g: \mathbb{Z} \rightarrow \mathbb{Z}$, $g(n) = n^2$

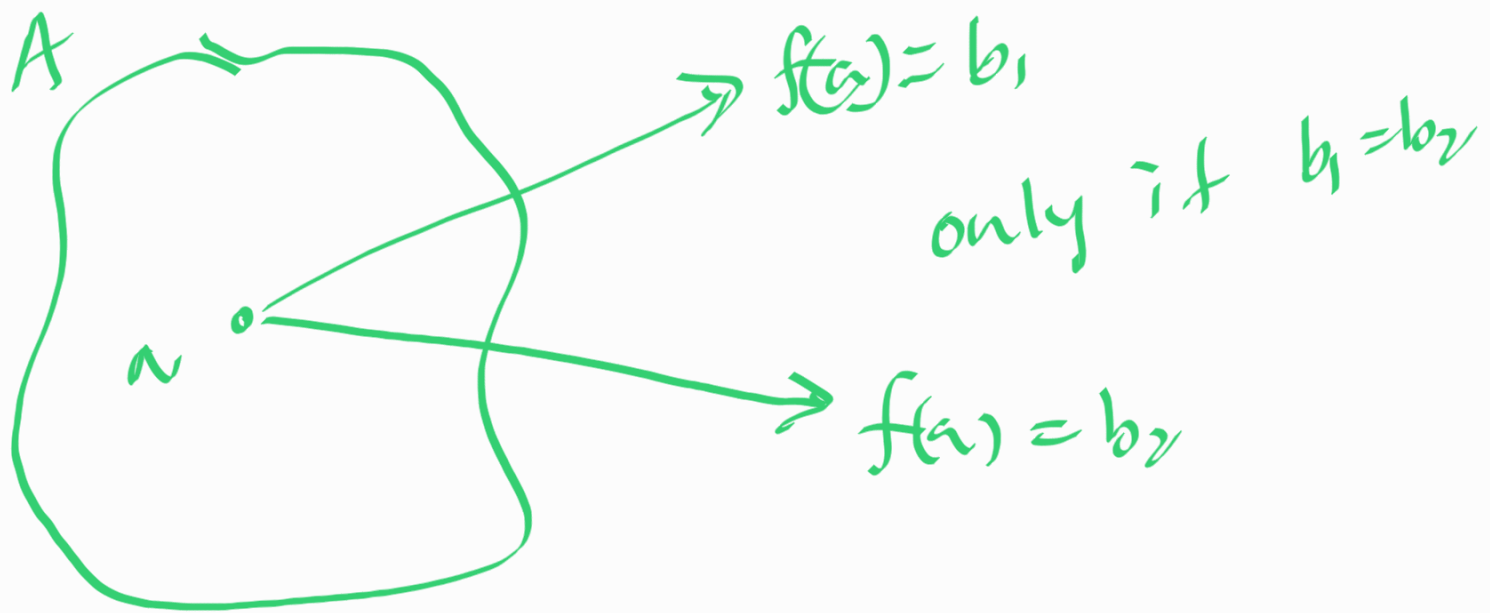
Equivalently, we can define
3 properties of a function.

suppose $f: A \rightarrow B$

① $\forall a \in A$, $f(a)$ is defined



② $\forall a \in A$, $f(a)$ must be unique.
In other words the function
 f must not generate different
values for a .



③ for each $a \in A$, $f(a) \in B$

The value generated by the function f should be an element in B .

$$\forall a \in A : f(a) \in B$$

Suppose you have a function

$$f: A \longrightarrow B$$

* A is called the domain of the function f .

* B is called the codomain of the function f .

* The range of the function f is the set of all elements that is generated by the function f .

$$\{ f(a) \in B : a \in A \}$$

ex: $f: \mathbb{Z} \rightarrow \mathbb{Z}$, f is
defined as $2n$
 $f(n) = 2n$

Domain: $\mathbb{Z}$

Codomain: $\mathbb{Z}$

Range: $\{x \in \mathbb{Z} : 2|x\}$

Range $\subseteq$ codomain.

- functions does not
necessarily need to have
real life idea. As long
as it maps all elements
in the domain to a unique
value in codomain we can
call it a function.

$$A = \{1, 2, 3\}$$

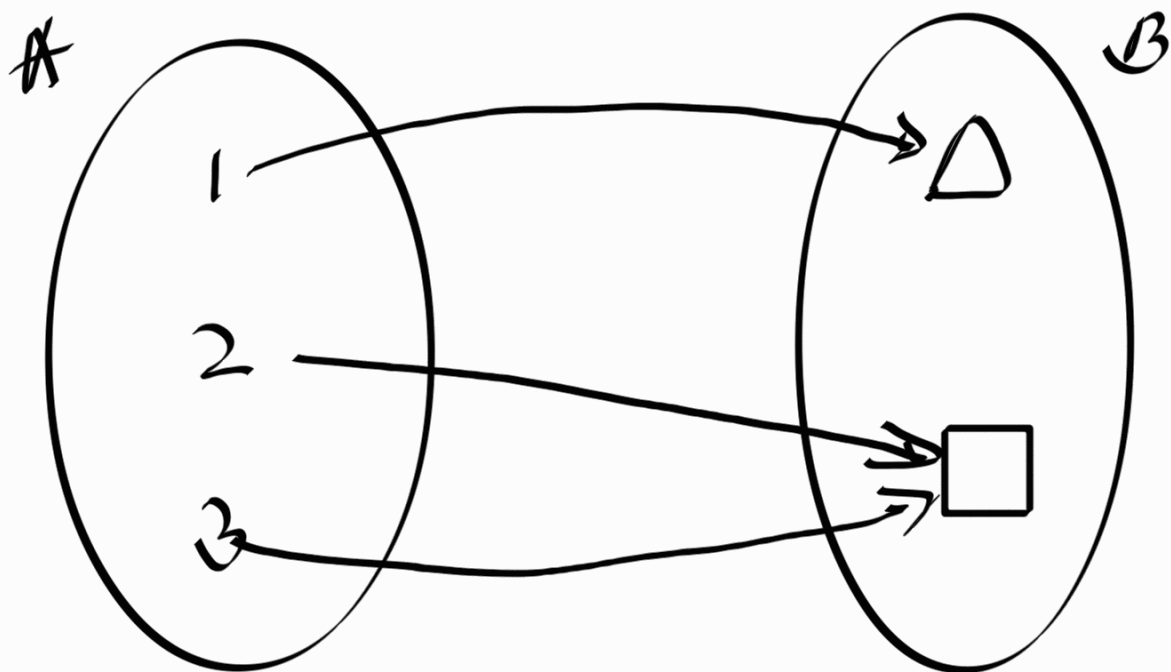
$$B = \{\triangle, \square\}$$

$a \in A$	$f(a)$
1	$\triangle = f(1)$
2	$\square = f(2)$
3	$\square = f(3)$

① $\forall a \in A: f(a)$
is defined

② $\forall a \in A: f(a)$
is unique

③ $\forall a \in A: f(a) \in B$



Ex: $f: \mathbb{R} \rightarrow \mathbb{R}$, defined as
 $f(x) = x^2$

Domain: $\mathbb{R}$

Codomain: $\mathbb{R}$

Range: $\mathbb{R}^{\geq 0}$

Rough proof idea $\rightarrow$

1. $\forall x \in \mathbb{R}$, $f(x) = x^2$, $f(x)$
is defined for all $x \in \mathbb{R}$.

2. $\forall x \in \mathbb{R}$, $f(x)$ is unique.

If $f(x) = a$ and $f(x) = b$

$$f(x) = a = x^2 \quad f(x) = b = x^2$$

$$a = b$$

∴ $f(x)$ is unique.

3. $\forall a \in A: f(a) \in \mathbb{R}$

Pop up quiz 5 answers

Q1: $g: \mathbb{Z} \rightarrow \mathbb{Z}, g(a) = 5$

1. $\forall a \in A: f(a)$ is defined
 $f(a) = 5,$

2. $\forall a \in A: f(a)$ is unique,
this is true because $\forall a \in A$
mapping only produces one
element which in this case
5.

$$3. \forall a \in A: f(a) \in \mathbb{Z}$$

this is true because

$$5 \in \mathbb{Z}$$



$$Q2: e: \mathbb{Z} \rightarrow \{T, F\}, e(x) = \begin{cases} T & \text{even} \\ F & \text{odd} \end{cases}$$

1. $\forall a \in \mathbb{Z}: e(a)$ is defined

all the elements in domain
is either even or odd,

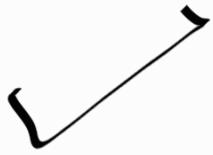
$\therefore e(a)$ is defined

2. $\forall a \in \mathbb{Z}: e(a)$ is unique,

all elements in $\mathbb{Z}$ is either
even or odd but not both,

3. $\forall a \in A: e(a) \in \{T, F\}$

$e(a)$ is always either
T or F.



(Q3) $p: \mathbb{Z}^{\geq 0} \rightarrow \mathbb{Z}^{\geq 0}$, defined
by

$$p(x) = x - 1$$

This is not a function
because it violates property
3.

when $x = 0$, $p(0) = 0 - 1 = -1$

$$-1 \notin \mathbb{Z}^{\geq 0}$$

Recap

- Def of a function

- $f: A \rightarrow B$

1. $\forall a \in A: f(a)$ is defined

2. $\forall a \in A: f(a)$ is unique

3. $\forall a \in A: f(a) \in B$

ex: $S: \mathbb{Z} \rightarrow \mathbb{Z}, S(x) = x+1$

Prove S is a function.

1. want to show (wts)

$\forall a \in \mathbb{Z}: S(a)$ is defined

$$S(a) = a+1$$

by the def of
 S .

2. WTS, $\forall a \in \mathbb{Z}$: $S(a)$ is unique,

If $S(a) = b_1$ $\wedge$ $S(a) = b_2$, then
 $b_1 = b_2$

Assume $S(a) = b_1$ and $S(a) = b_2$

$S(a) = b_1 = a+1$
 $S(a) = b_2 = a+1$ } by def of S and assumption.

$b_1 = b_2 = a+1$ by substitution

$$b_1 = b_2$$

$\therefore \forall a \in A$: $S(a)$ is unique.

$$3. \forall a \in \mathbb{Z} : s(a) \in \mathbb{Z}$$

Consider an arbitrary element
 $x \in \mathbb{Z}$

$$s(x) = x+1$$

by def of
 s

$$x+1 \in \mathbb{Z}$$

because $x \in \mathbb{Z}$
 sum of ints
 is an int.

s satisfies all 3 properties
 of a function.

$\therefore s$ is a function

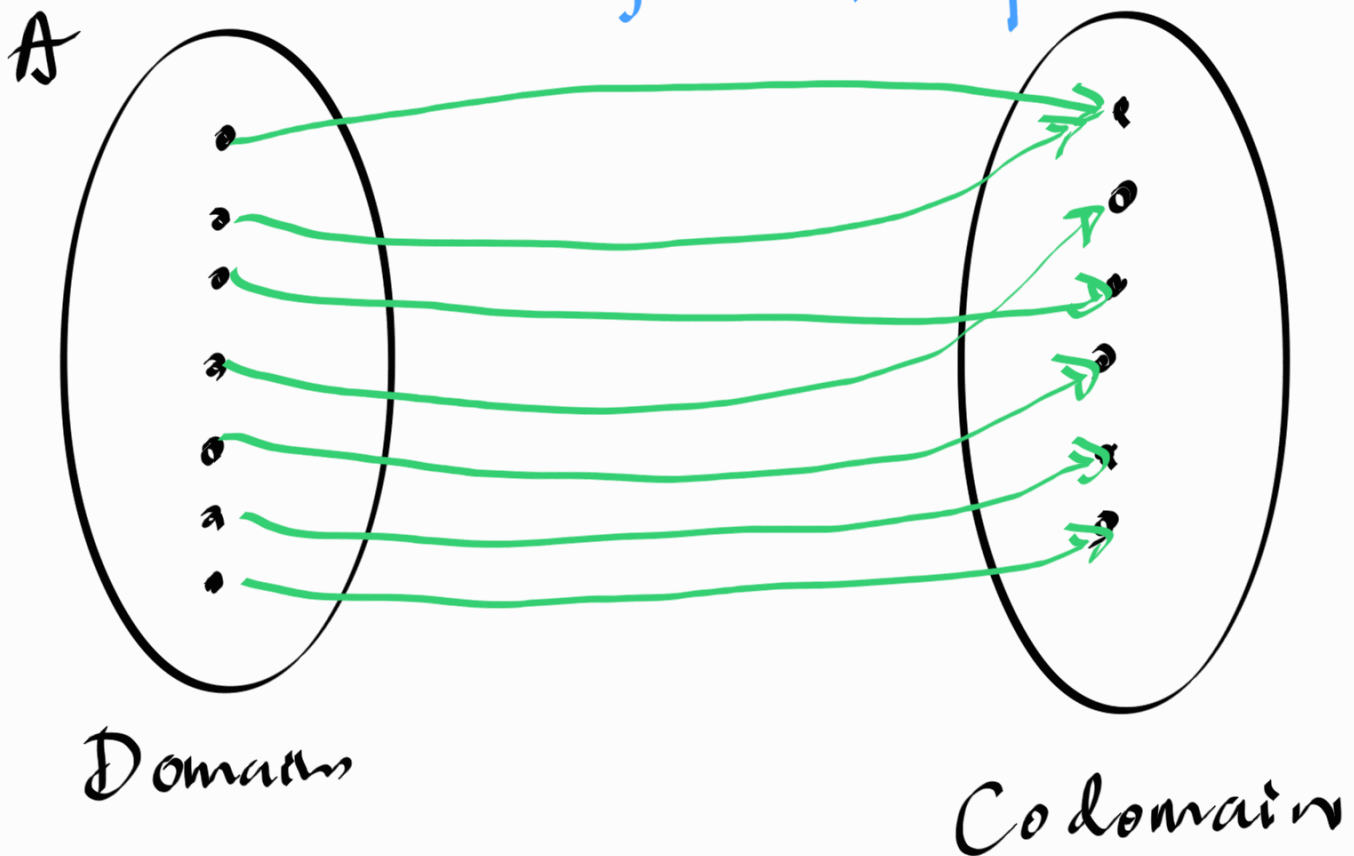


we are going to look at different types of functions.

Def onto/surjective functions

A function $f: A \rightarrow B$ is called onto (surjective)

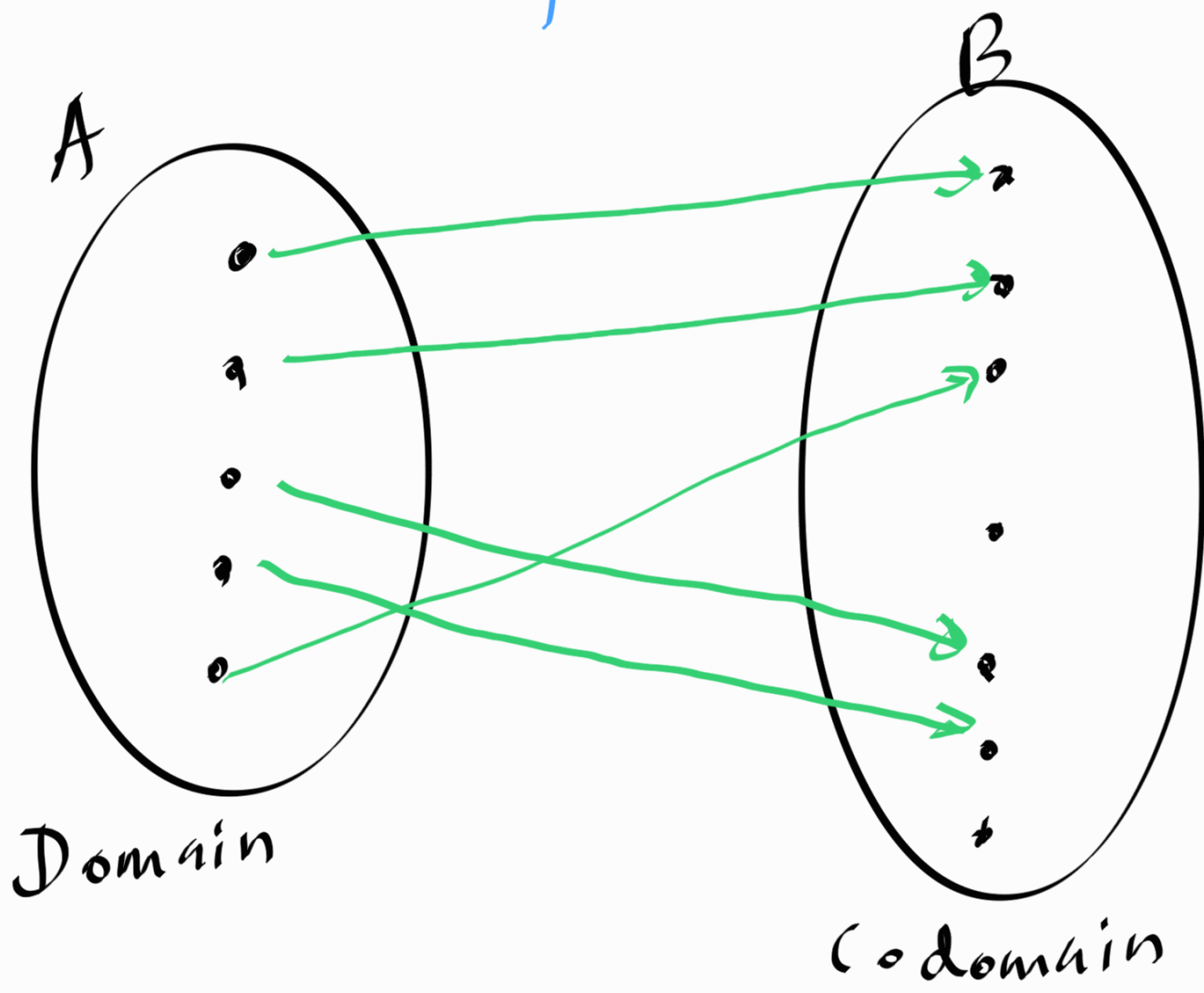
iff $\left[\forall b \in B : \left[\exists a \in A : f(a) = b \right] \right]$
 $\equiv \forall b \in B$, something in A maps ^{to} it.



Range = Co domain

Def one-to-one / injective functions,

A function $f: A \rightarrow B$ is called one-to-one (injective) iff $\forall a_1, a_2 \in A: a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2)$
 $\equiv \forall b \in B$, at most 1 element in A maps to it.



Injective and surjective
are NOT the opposite
of each other.

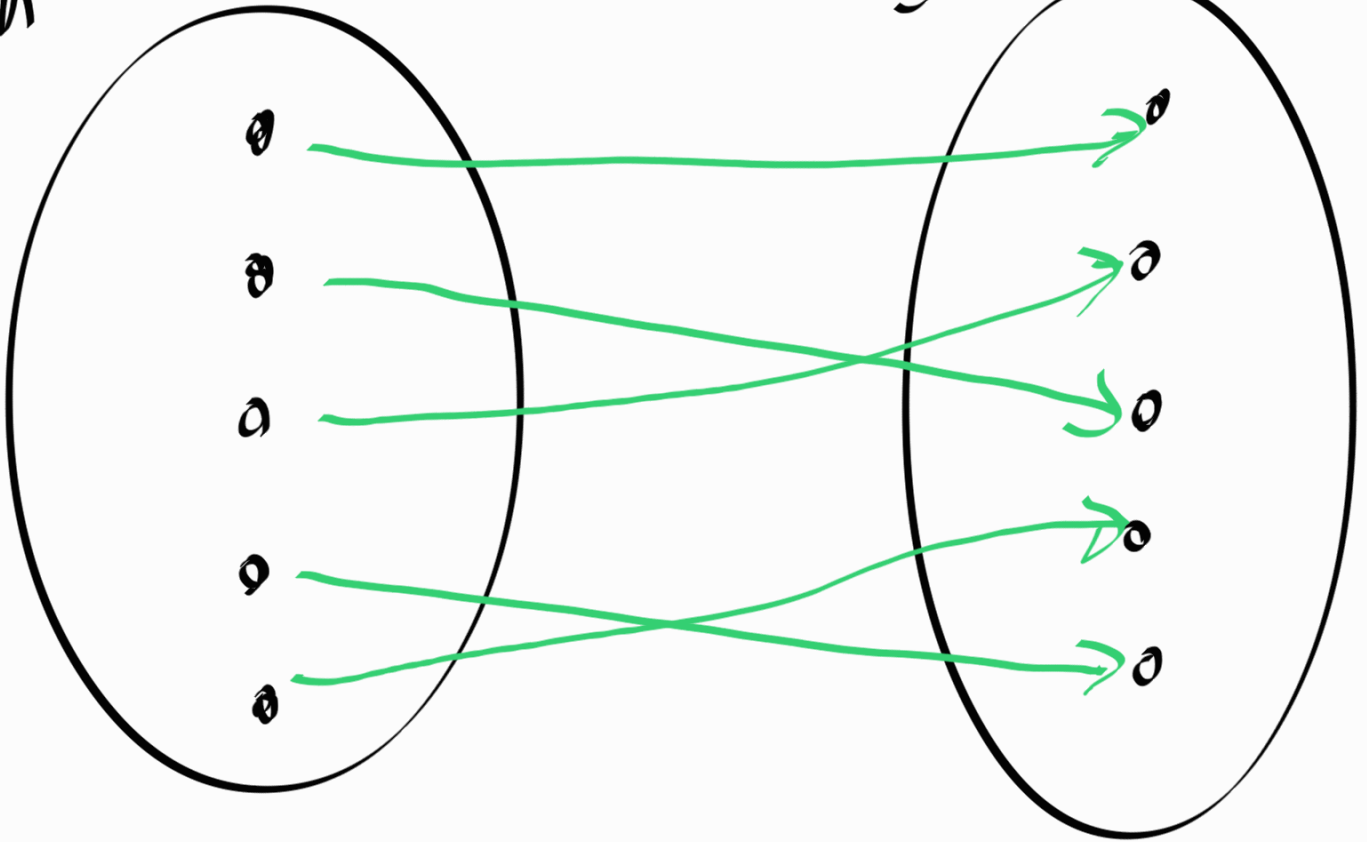
Def Bijection

A function $f: A \rightarrow B$
is called a bijection
if f is onto and
one-to-one.

$\equiv \forall b \in B$, exactly 1
element in A maps to
it.

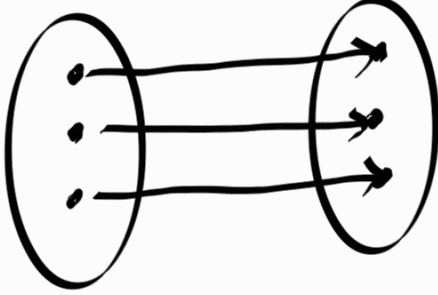
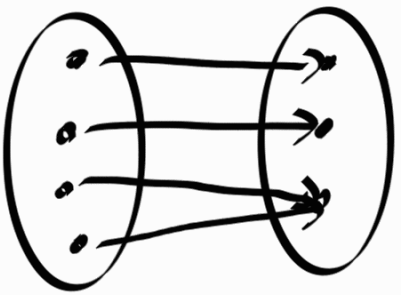
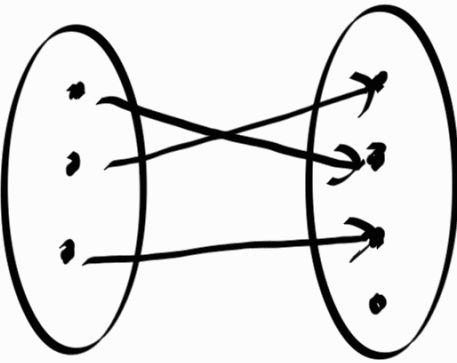
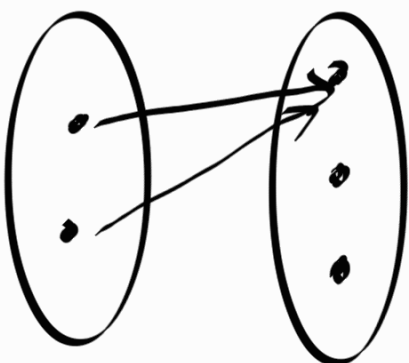
A

B



Domain

Codomain

	1:1 (one-to-one)	not 1:1
onto		
not onto		

How can we prove a function f is onto or 1:1?

onto

WTS

$$\left[\forall b \in B : [\exists a \in A : f(a) = b] \right]$$

$\equiv$

$$\text{If } b \in B \Rightarrow [\exists a \in A : f(a) = b]$$

example:

$$s: \mathbb{Z} \rightarrow \mathbb{Z}, \quad s(x) = x + 1$$

Let's prove that s is an onto function.

$$\text{WTS: } b \in \mathbb{Z} \Rightarrow [\exists a \in \mathbb{Z} : s(a) = b]$$

Proof

$$b \in \mathbb{Z}$$

by assumption

Let's construct $a \in \mathbb{Z}$
s.t. $S(a) = b$

Consider
 $a = b - 1$, $a \in \mathbb{Z}$

because $b \in \mathbb{Z}$

$$\text{now, } S(a) = b - 1 + 1$$

by def of S .

$$S(a) = b$$

So, we found $a \in \mathbb{Z}$ that maps to b .

$\therefore S$ is an onto function. $\square$

How to show a function $f: A \rightarrow B$ is not an onto function.

A function f is onto $\Leftrightarrow$
 $[\forall b \in B: \exists a \in A: f(a) = b]$

$$\neg [\forall b \in B: [\exists a \in A: f(a) = b]]$$

Note that:

$$\neg [\forall x \in S: P(x)] = [\exists x \in S: \neg P(x)]$$

$$\rightarrow [\exists b \in B: \neg [\exists a \in A: f(a) = b]]$$

Note that

$$\neg [\exists x \in S: P(x)] = [\forall x \in S: \neg P(x)]$$

$$[\exists b \in B : [\forall a \in A : \neg (f(a) = b)]]$$

$$[\exists b \in B : [\forall a \in A : f(a) \neq b]]$$